

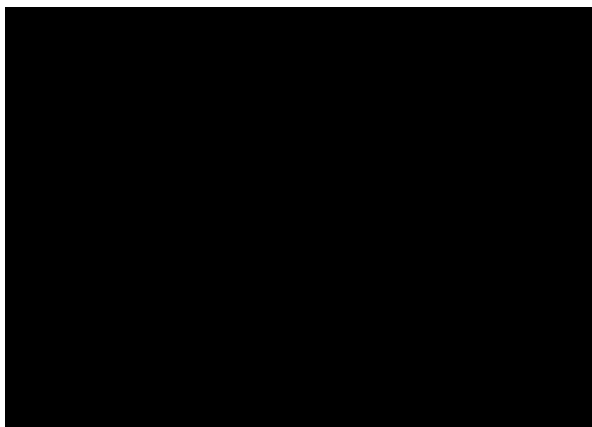
Individual Pathways

Activity 6.2

Investigating the atom

1. Insert a word to complete each sentence.
 - (a) John Dalton proposed that all matter consisted of tiny particles called.....
 - (b) J.J.Thomson proposed that atoms consisted of a sphere containing negatively charged electrons.
 - (c) Rutherford's alpha particle experiments revealed that atoms contained a central positive
 - (d) The nucleus of the atom contains protons and.....
 - (e) Bohr proposed that were located in fixed energy shells around the nucleus.

2. The following apparatus was used by Rutherford to investigate the structure of the atom.



- (a) Identify which of the following elements could be a source (X) of radiation: hydrogen, lead, gold, radium, carbon.
.....

- (b) X is protected by a box Y. Which of the following is a suitable material for box Y: carbon, gold, lead, copper.
.....

- (c) Identify the radiation labelled Z:

Activity 6.2

- (d) Most of the radiation strikes the screen without deflection. What does this imply about the atom?

.....

- (e) Explain why some radiation is deflected backwards at great angles.

.....

3. Complete the following table. You may need to refer to the periodic table.

Element	Symbol	Number of protons	Number of neutrons	Number of electrons
Fluorine	${}^{19}_{9}\text{F}$			
Potassium		19	21	
	${}^{239}_{94}\text{Pu}$			
Krypton		36	56	
	${}^{141}_{56}\text{Ba}$			

4. The model of the atom produced by Niels Bohr had electrons arranged in energy levels or shells at different radial distances from the nucleus. The first shell ($n=1$; closest to the nucleus) was called the K shell. The table shows the shell number and name. Given that the maximum number of electrons in each shell 'n' is $2n^2$, complete the table. The first row has been completed for you.

Shell number (n)	Shell letter	Maximum electron number
1	K	$2(1)^2 = 2$
2	L	
3	M	
4	N	
5	O	

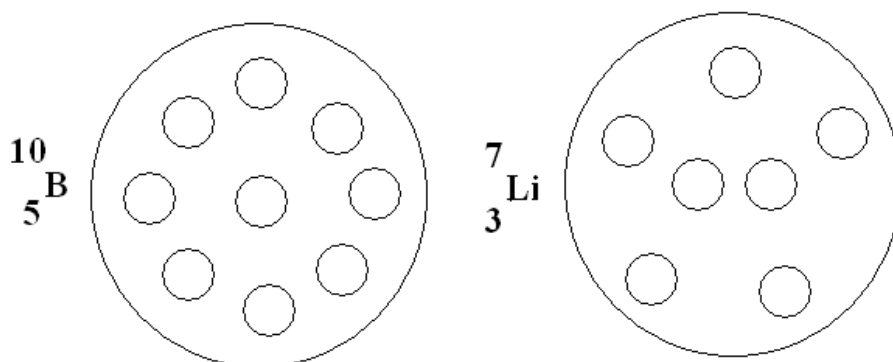
5. Each chemical element has a different number of protons and electrons. Electrons fill the electron shells starting from the K shell and working outwards. Once a shell is filled, any remaining electrons start to fill the next shell. Thus beryllium (Be) has four electrons and the first two electrons fill the K shell. The remaining two electrons are placed in the L shell.

Activity 6.2

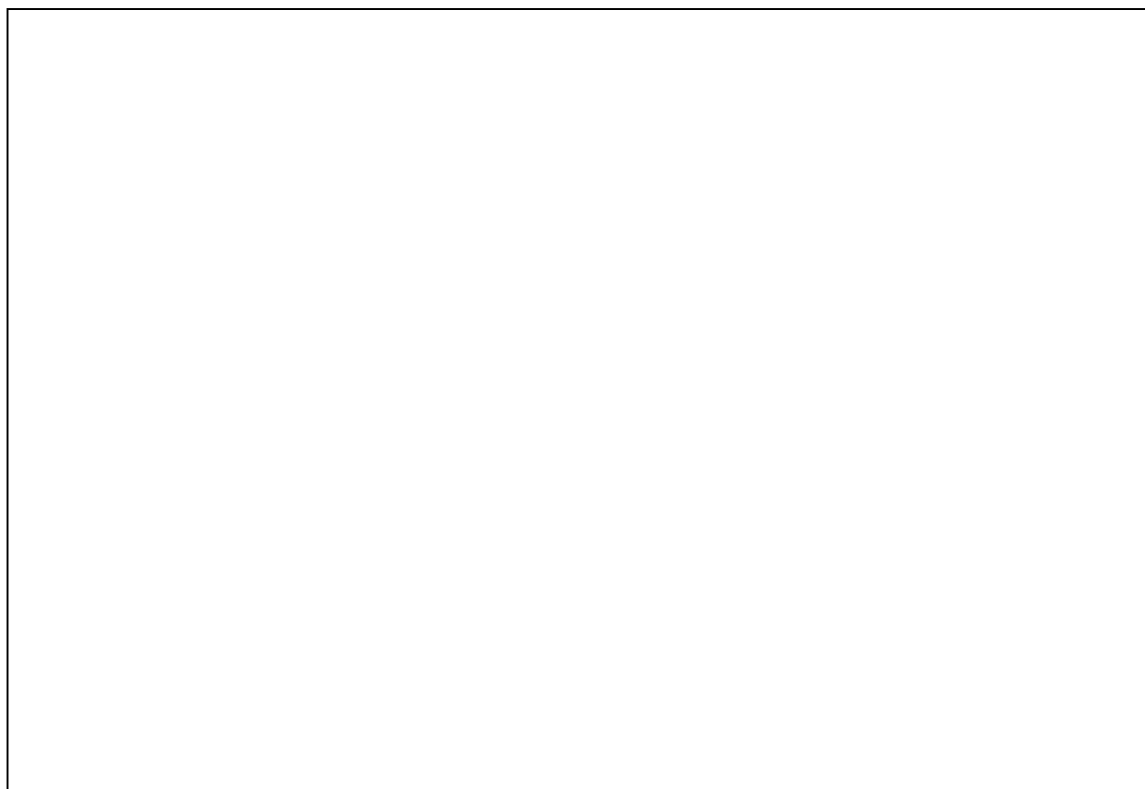
Based on this information, complete the following table.

Element	Z	K shell	L shell	M shell
Carbon	6			
Sodium	11			
Silicon	14			
Chlorine	17			

6. The following diagram shows the nuclei of various atoms. Each circle represents a proton or a neutron. In each example, shade in the neutrons that are present.

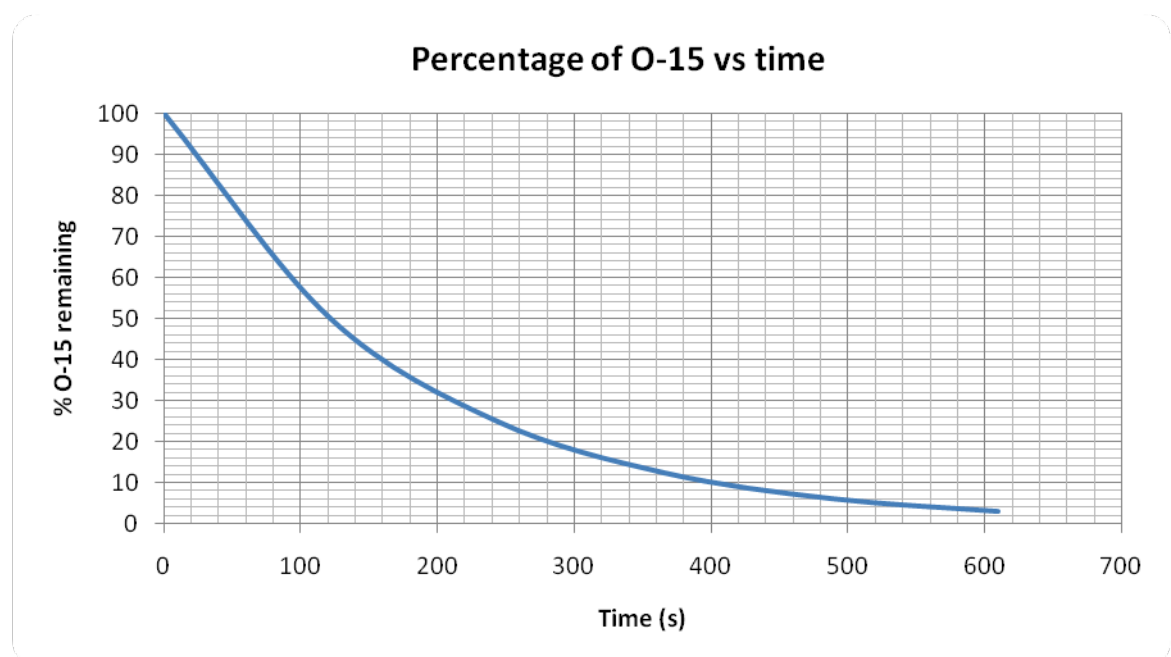


7. Lead-211 has a half-life of 36 minutes. A 50 g sample of Pb-211 was obtained and the mass remaining measured as a function of time. Construct a line graph of the mass of lead-211 remaining after six half-lives have elapsed.



Activity 6.2

8. The following graph shows the decay curve for oxygen-15.



Using the graph, estimate the half-life of oxygen-15.

9. Bismuth-210 ($Z=83$) radioactively decays to form polonium-210 ($Z=84$).

(a) Calculate the number of protons and neutrons in the bismuth-210 isotope.

.....

(b) Calculate the number of protons and neutrons in the polonium-210 isotope.

.....

(c) Explain what has happened during this nuclear decay.

.....

10. In the natural decay of U-238 ($Z=92$), the daughter nucleus formed is Th-234 ($Z=90$). Determine the type of radiation emitted.

.....

Answers

1.

- (a) John Dalton proposed that all matter consisted of tiny particles called **atoms**.
- (b) J.J.Thomson proposed that atoms consisted of a **positive** sphere containing negatively charged electrons.
- (c) Rutherford's alpha particle experiments revealed that atoms contained a central positive **nucleus**.
- (d) The nucleus of the atom contains protons and **neutrons**.
- (e) Bohr proposed that **electrons** were located in fixed energy shells around the nucleus.

2.

- (a) X = radium
- (b) Y = lead
- (c) Z = alpha particles
- (d) The atom is largely empty space
- (e) Some of the alpha particles are on (or almost on) target for the tiny positive nucleus. They are deflected backwards by electrostatic repulsion.

3.

Element	Symbol	Number of protons	Number of neutrons	Number of electrons
Fluorine	${}^{19}_{9}\text{F}$	9	10	9
Potassium	${}^{40}_{19}\text{K}$	19	21	19
Plutonium	${}^{239}_{94}\text{Pu}$	94	145	94
Krypton	${}^{92}_{36}\text{Kr}$	36	56	36
Barium	${}^{141}_{56}\text{Ba}$	56	85	56

Activity 6.2

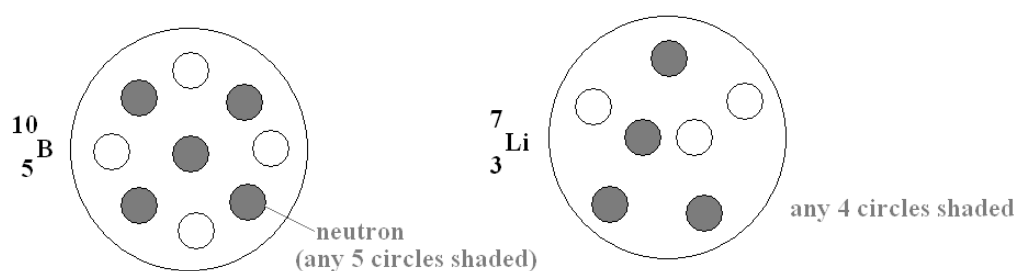
4.

Shell number (n)	Shell letter	Maximum electron number
1	K	$2(1)^2 = 2$
2	L	$2(2)^2 = 8$
3	M	$2(3)^2 = 18$
4	N	$2(4)^2 = 32$
5	O	$2(5)^2 = 50$

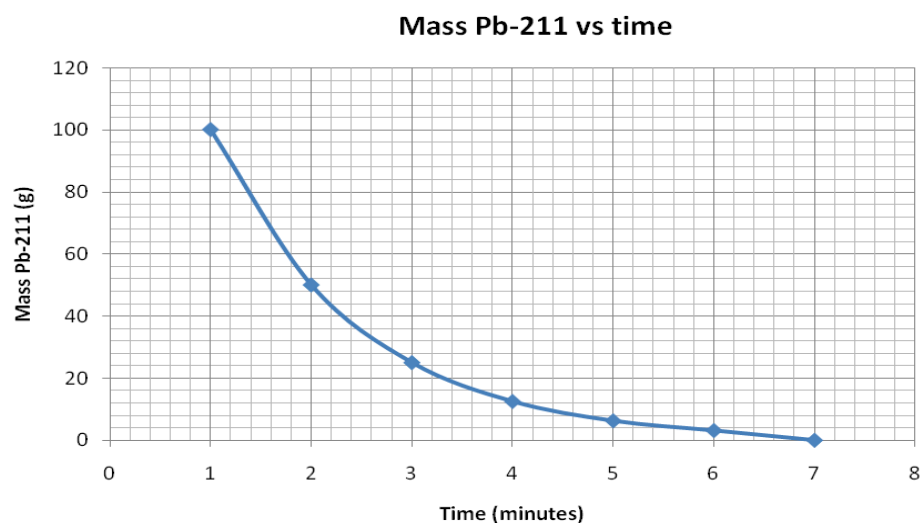
5.

Element	Z	K shell	L shell	M shell
Carbon	6	2	4	
Sodium	11	2	8	1
Silicon	14	2	8	4
Chlorine	17	2	8	7

6.



7.



8. Half-life = 120 seconds

9.

(a) $p = 83$

$$n = 210 - 83 = 127$$

(b) $p = 84$

$$n = 126$$

(c) One neutron has decayed and a proton has formed. To maintain a charge balance, a beta particle must also have formed and been ejected as radiation.

10. U-238 has 92 protons and 146 neutrons. Th-234 has 90 protons and 144 neutrons. Thus the radiation released has 2 protons and 2 neutrons. This is an alpha particle.